

AI Is Here To Stay: Misinformation and Human-Centric Models Between Risks and Opportunities

MARCO BELLÒ, University of Padua, Italy

For more than half a century, artificial intelligence remained a matter of speculation for novelists and philosophers. This changed in 2017 due to the introduction of the transformer architecture, spearheading the AI boom whose effects are still scarcely understood. This paper explores potential risks and use cases of this technology, with a "human-centric" cut: the first half showcases generation and detection of fake news and deepfakes, as well as how much LLMs can influence people's opinions and beliefs. The focus then shifts to the AIs themselves, about their own biases and their susceptibility to external stimuli, concluding with a quick overview on the state of research about AIs for positive social impact.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Artificial intelligence**; **Natural language processing**; **Reasoning about belief and knowledge**; Cognitive science.

Additional Key Words and Phrases: AI, NLP, Misinformation, Deepfakes, Opinions, Biases, AI4SI

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1 INTRODUCTION

Politicians are some of the most influential people in human society: from the greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to steer society. Shapiro and Page in 1984 showed how nearly all 20th century U.S. Presidents have tried to lead public opinion, with varying degrees of success. [13]. As societies became more democratic, so grew the leaders' responsibility to convince potential followers that their policies are beneficial. Thus, speeches play a vital role in the functioning of society [2]: leaders depend on the verbal power to persuade people [4].

Moreover, democratic stability depends on citizens on the losing side accepting election outcomes. Clayton et. al. evaluated the effects of exposure to multiple statements from (at the time) former president Donald Trump attacking the legitimacy of the 2020 US presidential election. They found no evidence indicating that elites can erode democratic norms easily or that the effects of norm violations are uniform across the entire population. However, elite rhetoric can shape normative beliefs in core democratic values such as confidence in elections and support for peaceful transfers of power. This was especially evident amongst people who approved of Trump's performance in office [6].

Having established the importance of speeches, we can differentiate between two types: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social networks posts. This last category is especially interesting since it affords politicians a

Author's address: Marco Bellò, marco.bello.3@studenti.unipd.it, University of Padua, Padua, Italy.

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direct and high-volume communication channel to their potential followers: U.S. President Donald Trump tweeted 57.0000 times between May 2009 and January 2021 [10] alone, while Italy's Prime Minister Giorgia Meloni totalled more than eighty million social networks interactions in 2025 [16].

On the other hand, internal speeches are especially easy to access, being already aggregated in public databases, yet their technical and codified nature makes their interpretation harder for both human and automated agents. Moreover, until 1985 Italian stenographic documentation did not report speeches word for word, but rather translated into a "language of the parliament" [8, 12].

2 SOCIAL NETWORKS

Broadly speaking, the goal of this project is to gather a corpus of politicians' speeches and perform linguistic analyses on it. The idea is to start from standard NLP tasks such as word frequency and text complexity, to then move to classification tasks such as hate speech recognition, and finally to fine-tuning a LLM.

This objective is why I ultimately decided to gather a corpus of *external speeches* rather than internals. Large language models are trained with tokenizers, and the resulting token distribution is highly imbalanced: Chung et. al. did a controlled study that scales the vocabulary of the language model from 24K to 196K while holding data, computation, and optimization unchanged. They discovered that models are disproportionately optimized on the part of language that appears most often, meaning they are way better at understanding common words and patterns [5]. This translates to the fact that internal speeches are less understandable for an AI agent.

As previously stated, external speeches include social networks communications. While television is still the first source of information for the majority of the population, online platforms are steadily growing and are already the preferred media for young people [7, 11]. There are evidence of increase in political participation due to social media usage, as well as risks for the functioning of democracy [1, 9], and there have been attempts to predict election results with social media data analyses. Rita et. al. measures sentiment polarity on Twitter, and concludes that tweets' sentiment is not a reliable election results predictor. Additionally, results also show that it is impossible to state that social media impacts voting decisions [14].

Opposite results can be seen in Belcastro et. al. 2022 study: a real-time analysis was carried out during the 2020 US presidential election campaign correctly identifying the leading candidate before the Election Day in 10 out of 11 swing states [3]. Similar results can be seen in Brito et. al.

Silva et. al. managed to match tweets against parliamentary speeches to measure politicians sentiment on a specific topic. They find that parliament members who participate less in parliamentary debate tend to have larger differences with their party on Twitter, suggesting that a certain level of self-censoring is taking place in the parliamentary arena [15].

Lastly, social networks opens the possibility of semi-automatized corpora building.

3 PROJECT GOAL

Lets now define the project's goal.

I want to compare politicians speeches among three countries, namely Italy, France and the United States, between 1945 and 2025, and then cross-compare the resulting analyses with democracy level indicators taken from the V-Dem dataset¹.

¹The V-Dem Dataset is a multidimensional and disaggregated dataset that reflects the complexity of the concept of democracy as a system of rule that goes beyond the simple presence of elections: <https://www.v-dem.net/data/the-v-dem-dataset/>.

I have chosen these three countries for several reasons:

The first step is building a corpora of politicians speeches, one corpus for italian speeches, one for french speeches and one for american speeches.

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